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Abstract—Massive activations, characterized by exceptionally high activation values within large language models, pose significant challenges to numerical stability and computational efficiency. This paper presents a systematic investigation into the generation mechanisms of massive activations across diverse transformer-based architectures. Through a series of five systematically designed experiments, we demonstrate that massive activations originate from multi-layer perceptron (MLP) layers rather than attention mechanisms. Our analysis reveals that function words serve as key triggers and that activation patterns exhibit strong alignment with the dominant singular vectors of MLP weight matrices via Singular Value Decomposition (SVD). V-matrix ablation experiments provide causal evidence supporting the SVD-based amplification mechanism. These findings offer novel insights for improving model stability and interpretability, with implications for model optimization and deployment. Code is available at: https://github.com/Ludan-daye/changeHead_massvieActive.

Index Terms—Massive Activations, Large Language Models, Function Words, SVD, MLP, Geometric Interpretation.

I. INTRODUCTION

Large language models (LLMs) based on the Transformer architecture [1] have achieved unprecedented success across diverse domains of natural language processing [2], [3]. This success is largely attributed to scaling laws, where model performance predictably improves with increases in parameters and training data [4]. However, this rapid growth in scale and complexity has rendered the internal computations of models increasingly opaque, creating significant challenges for ensuring their robustness, reliability, and interpretability [5], [6].

Within this context, the phenomenon of **Massive Activation** (MA) has emerged as a critical area of concern [7]. MA is characterized by activation values orders of magnitude larger than the typical range [8], posing a direct threat to numerical stability, especially in low-precision inference settings [9]. The

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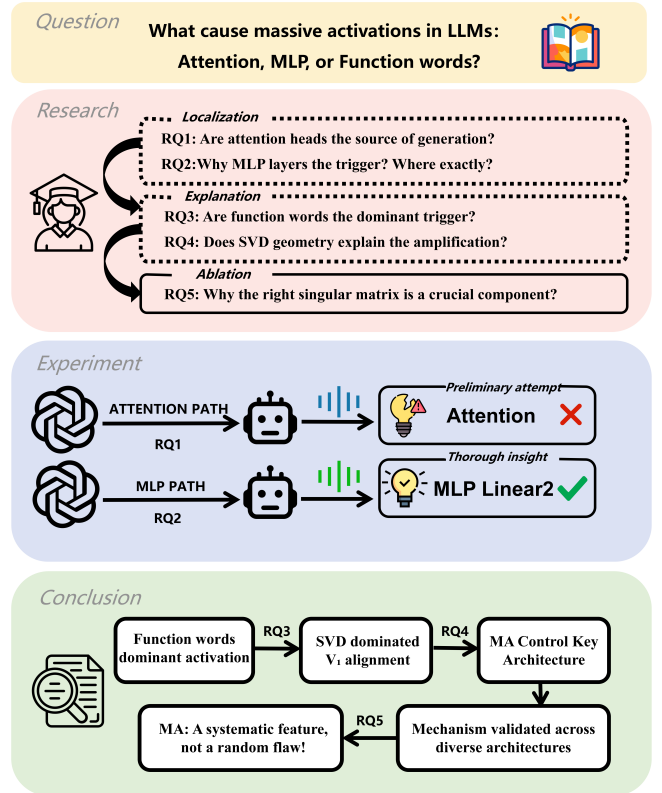


Fig. 1: Research Framework and Process

trend towards sparsely activated, mixture-of-experts models further amplifies the need to understand such internal dynamics [10].

While the existence of massive activations is known [7], research into their root causes remains limited. Prevailing explanations often attribute them to broad architectural or training instability, overlooking the potential interplay between

linguistic structure and the **geometric properties** of model weights [11], [12]. A key gap is the lack of evidence on the role of **function words** as triggers [13], and more importantly, a systematic causal framework linking trigger, internal computation, and amplification [14].

To bridge this gap, our study employs a mechanistic interpretability approach [15]. We use Singular Value Decomposition (SVD) to analyze the geometry of weight matrices [16], [17] and apply causal intervention methods to establish causal relationships [18], [19]. Guided by formal causal reasoning [20], our investigation is structured around five targeted research questions, as illustrated in the research framework (Fig. 1):

- RQ1:** *Source Effect.* Does massive activation originate from the attention mechanism or the MLP module?
- RQ2:** *Localization Effect.* If it originates from the MLP, which internal component plays the dominant role?
- RQ3:** *Trigger Effect.* Is the phenomenon triggered by specific types of vocabulary, such as function words?
- RQ4:** *Mechanism Effect.* Can the geometric structure of weight matrices (SVD) explain its generation mechanism?
- RQ5:** *Causality Effect.* Does this geometric structure have a causal determining force on massive activation?

Our work provides an end-to-end elucidation of the **“function word triggering – MLP generation – SVD geometric amplification”** mechanism. Our analysis covers eight English-language dense Transformer models ranging from 124M to 13B parameters, spanning three normalization schemes (Post-LN, Pre-LN, RMSNorm) and three activation functions (ReLU, GELU, SwiGLU). Since the identified geometric mechanism operates on weight-space spectral structure rather than language-specific lexical features, we expect it to generalize beyond English; verification across other languages, larger scales (>70B), and sparse architectures (e.g., Mixture-of-Experts) is left for future work. This insight offers a precise target for future mitigation strategies in model architecture [21], training [22], [23], and efficient inference [24], contributing to more robust and interpretable LLMs.

II. RELATED WORK

Our investigation into the mechanism of massive activations intersects with several established lines of research in mechanistic interpretability and analysis of language models. This section reviews the pertinent literature, structured around four core themes that directly inform our research questions and methodological approach.

a) Massive Activations (MA): The phenomenon of exceptionally large activation values within neural networks has been documented as a challenge for model stability and efficiency. Recent work by [7] systematically identified and

named this “massive activation” phenomenon in large language models, highlighting its correlation with numerical instability during low-precision inference. Earlier, [8] discussed the saturation limits of Transformer activations in a formal computational framework, linking extreme values to potential limitations in expressivity. Research on improving hardware resilience, such as the algorithm-based error tolerance approach in [9], further underscores the practical importance of understanding and mitigating such numerical anomalies. Beyond immediate stability, the persistence of abnormal activation patterns may also relate to broader model behavioral issues, such as the inefficient utilization of long contexts [26] and challenges in post-training quantization [27]. However, these studies primarily treat MA as a symptomatic problem to be circumvented or mitigated at the system level [25], rather than probing its origin within the model’s computational graph and its intrinsic link to linguistic processing. Our work moves beyond this symptomatic view, aiming to decompose the causal pathway of MA generation.

b) The Role of the MLP in Transformers: The Multi-Layer Perceptron (MLP) blocks in Transformer models are increasingly recognized as primary sites for non-linear feature transformation and knowledge application. Seminal work by [?] demonstrated that feed-forward layers function as key-value associative memories, where specific neurons activate for distinct semantic or syntactic concepts. This view is complemented by the “circuit” perspective introduced by [14], which provides a mathematical framework for decomposing Transformer computations into subgraphs, often highlighting the MLP as a critical component in factual recall and pattern completion. Subsequent research has further identified “knowledge neurons” within MLPs [28] and sought to scale up the identification of causal mechanisms [29]. While the functional specialization of MLPs is well-established, their specific role in generating *pathological* or extreme activations, as opposed to normal signal amplification, remains underexplored. Furthermore, most editing and localization techniques focus on semantic knowledge [30], leaving a gap in understanding purely syntactic triggers. Our research (RQ1 and RQ2) directly addresses this gap by experimentally isolating the MLP—and specifically the `down_proj` layer—as the physical source of massive activations, distinguishing its function from that of the attention mechanism.

c) Function Words as Linguistic Triggers: The differential processing of function words (e.g., prepositions, conjunctions) versus content words by neural models has been a subject of interest in interpretability. Probing studies have consistently shown that syntactic information, often carried by function words, is reliably encoded in the representations of models like BERT [11], [12]. Further analysis by [13] revealed that BERT captures hierarchical linguistic structures, with certain layers specializing in syntactic features. Research on attention heads also suggests specialization for syntactic roles [31], and syntax-aware modeling highlights the importance of these structures [32]. More recent fine-grained analyses continue to explore how LLMs handle various linguistic phe-

nomena [33]. However, this body of work primarily establishes correlation—showing that representations *contain* syntactic information. It does not investigate whether these specific linguistic categories can act as *triggers* for extreme computational states, such as massive activations. Our study (RQ3) bridges this divide by empirically testing and confirming the hypothesis that function words are the dominant linguistic trigger for MA, thereby linking a well-studied representational property to a specific, impactful dynamical behavior within the model.

d) *Geometric Analysis via SVD and Causal Intervention*: Analyzing the geometric structure of neural networks via Singular Value Decomposition (SVD) has proven fruitful for model compression and interpretation. [16] showed that the singular vectors of Transformer weight matrices correlate with human-interpretable linguistic concepts. Building on this, [17] proposed activation-aware SVD for more efficient model compression, and spectral analysis has been applied to understand internal representations more broadly [34], [35]. To move from observing geometric correlations to establishing causal mechanisms, the field of mechanistic interpretability has developed interventionist techniques. Work such as [18] on locating and editing factual knowledge and [19] on identifying important directions in activation spaces exemplify this paradigm. Recent advances continue to refine causal abstraction methods [36], knowledge removal techniques [37], and geometric perspectives on robustness [38]. These approaches are grounded in a formal causal reasoning framework [20]. Our work (RQ4 and RQ5) synthesizes these two strands: we employ SVD to identify the geometric alignment between function word inputs and weight matrix singular vectors as the putative amplification mechanism, and then design a novel *V-matrix ablation* experiment—a direct causal intervention on the model’s geometry—to test this hypothesis, thereby providing a causal explanation rooted in the model’s linear algebraic structure. This also opens avenues for future efficient inference methods inspired by dynamic activation patterns [39].

III. PROBLEM FORMULATION AND EXPERIMENTAL SETUP

To systematically investigate the causes of massive activations, we adopt a multi-stage experimental framework, as outlined in Fig. 2, which connects our hypothesis testing with causal verification. We begin by formally defining the key concepts and evaluation metrics, followed by implementation details. Each subsequent section then presents the methodology and results for one research question in an integrated manner.

A. Metrics and Definitions

Let $\mathbf{A} \in \mathbb{R}^{B \times L \times D}$ denote an internal activation tensor, where B , L , and D represent batch size, sequence length, and hidden dimension, respectively.

Definition 1 (Massive Activation). An activation value $a \in \mathbf{A}$ is classified as a *massive activation* (MA), as visually exemplified in Fig. 3, if its magnitude exceeds a statistical threshold

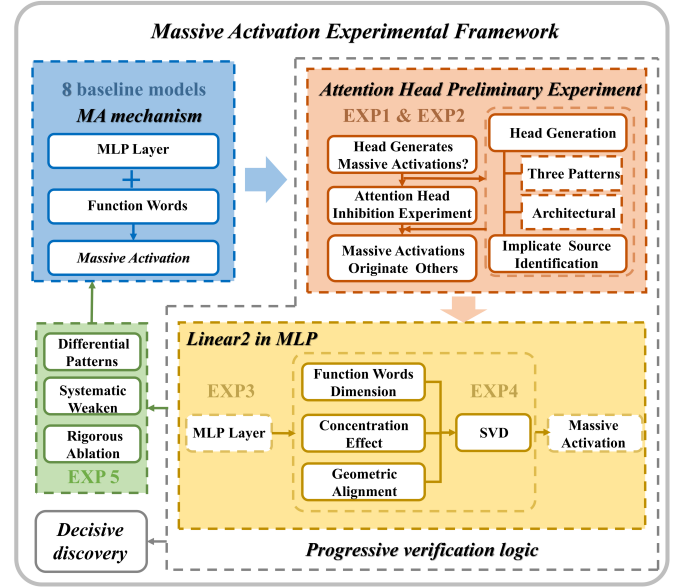


Fig. 2: Experimental Framework

T , defined as the 99.9th percentile of the absolute activation distribution:

$$\mathcal{M} = \{a \in \mathbf{A} \mid |a| > T\}, \quad \text{where } T = P_{0.999}(|\mathbf{A}|). \quad (1)$$

Here, $P_p(\cdot)$ denotes the p -th percentile operator.

Definition 2 (Top1 Activation). The maximum absolute activation value across all network positions serves as our primary intensity metric:

$$\text{Top1} = \max_{(b,l,d) \in \mathcal{I}} |A_{b,l,d}|, \quad (2)$$

where $\mathcal{I} = \{1, \dots, B\} \times \{1, \dots, L\} \times \{1, \dots, D\}$ is the index set.

Definition 3 (ΔTop1 Change Rate). For intervention experiments, we quantify the relative change in activation intensity as:

$$\Delta\text{Top1} = \frac{\text{Top1}^{(\text{int})} - \text{Top1}^{(\text{base})}}{\text{Top1}^{(\text{base})}} \times 100\%, \quad (3)$$

where superscripts (base) and (int) denote baseline and intervention conditions, respectively.

Definition 4 (SVD Alignment). Let $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be a weight matrix with singular value decomposition $\mathbf{W} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_{\min(d_{\text{out}}, d_{\text{in}})})$ with $\sigma_1 \geq \dots \geq \sigma_{\min(d_{\text{out}}, d_{\text{in}})} \geq 0$. Given an MA direction vector $\mathbf{a} \in \mathbb{R}^{d_{\text{in}}}$, its alignment with the i -th right singular vector $\mathbf{v}_i$ (the i -th column of $\mathbf{V}$) is defined as:

$$\varrho(\mathbf{a}, \mathbf{v}_i) = \frac{\mathbf{a}^T \mathbf{v}_i}{\|\mathbf{a}\|_2 \cdot \|\mathbf{v}_i\|_2} \in [-1, 1]. \quad (4)$$

A value of $|\varrho| \approx 1$ indicates strong directional alignment.

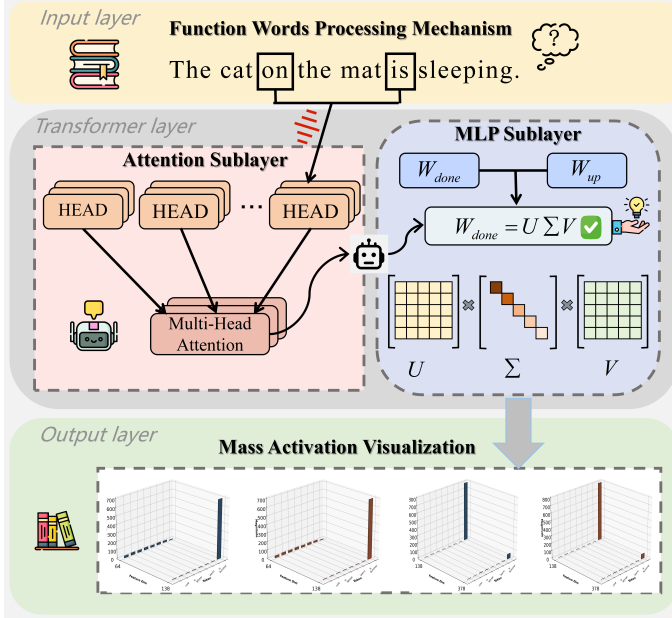


Fig. 3: Massive Activation

B. Models, Data, and Statistical Methods

Models: All experiments are conducted across eight representative LLMs: GPT-2 (124M), LLaMA-2-13B, LLaMA-2-7B-Chat, BLOOM-7B1, GPT-J-6B, QWEN2.5-7B, OPT-6.7B, Falcon-7B, and Mistral-7B-v03, loaded from the Hugging Face Transformers library using standard configurations.

Data: We sample 64 documents independently from the C4 validation split (English, allenai/c4) using a fixed random seed (42), retaining only documents whose tokenized length is at least 256 BPE tokens. From each selected document we extract a single contiguous window of the first 256 tokens (no concatenation across documents), yielding 64 statistically independent observations per model. This protocol replaces the RedPajama-1T-Sample workflow used in the initial submission—both because that dataset is no longer hosted on the Hugging Face Hub and to ensure that every analysis window comes from a distinct source document, directly addressing the sample-dependency concerns raised in review.

To verify that this change does not affect the central conclusions, we performed two robustness checks under the new protocol across seven models (LLaMA-2-13B is excluded due to a tokenizer caching issue). First, we re-ran the function-word trigger analysis (RQ3): across all MA layers (top-1/median ratio $> 30\times$), function words account for 40.6–53.1% of top-1 activation tokens (GPT-2: 53.1%, Qwen2.5: 40.6%, GPT-J: 51.6%, Mistral: 41.0%, BLOOM: 45.3%, Falcon: 51.6%), substantially exceeding the $\sim 30\%$ base rate of function words in natural text. Second, we re-ran the V -matrix ablation (RQ5) on models whose key layer matches Table I: GPT-2 ($\Delta MA = -75.9\%$, paper: -70.7% , $\delta = 5.2$ pp), Qwen2.5-7B (-99.1% , $\delta = 0.0$ pp), and GPT-J-6B (-68.4% , paper: -69.6% , $\delta = 1.2$ pp). Both checks

confirm that the core findings—function words preferentially trigger MAs, and V -matrix geometry is causally necessary—are robust to independent document-level sampling.

Statistical Analysis: For comparative experiments (RQ1, RQ3, RQ5), we employ paired t-tests with significance threshold $p < 0.01$. Effect sizes are reported using Cohen’s d where applicable. Correlation analyses use standard correlation coefficients with confidence intervals.

IV. EXPERIMENTS AND RESULTS

This section presents each research question with its methodology and results integrated, progressing from source identification (RQ1–RQ2) through trigger analysis (RQ3) to geometric mechanism discovery (RQ4) and causal verification (RQ5).

A. (RQ1) Attention Mechanism Contribution Analysis

Hypothesis and Method. We test whether attention mechanisms directly generate MAs or merely modulate input patterns to downstream modules. For each model, we perform two inference passes: (1) *Baseline*: normal forward pass; (2) *Intervention*: all attention head outputs are dynamically zeroed via PyTorch hooks while preserving other components. We compare ΔTop1 between conditions. A negative ΔTop1 indicates attention promotes MA generation, while a positive value suggests suppression.

Results.

a. Three distinct regulatory patterns. Table I reveals three clear patterns in how attention mechanisms influence massive activation generation. *Generative-promoting models* (GPT-2, LLaMA-2, BLOOM, GPT-J) show significant negative ΔTop1 values (-60% to -98%), indicating attention actively contributes to MA generation. *Inhibitory-regulatory models* (Qwen2.5-7B, OPT-6.7B) exhibit dramatic positive ΔTop1 values ($> +250\%$), demonstrating attention’s role as a suppressor. *Hybrid models* show complex behaviors: Mistral-7B-v03 displays layer-specific divergence, while Falcon-7B exhibits auxiliary characteristics.

b. Architectural evolution and consistency. The results trace an evolutionary trajectory from classical to modern architectures. GPT-2 (-60%) establishes the baseline generative pattern, consistently maintained in its architectural descendants (LLaMA-2-13B: -79.5%). The extreme suppression in BLOOM-7B1 (-98.3%) suggests ALiBi positional encoding amplifies attention’s generative role. The inversion observed in Qwen and OPT represents a notable design shift, where attention mechanisms evolve from generators to stabilizers. Mistral’s layer-specific behavior indicates sophisticated internal computation management.

c. Implications for source identification. The diverse ΔTop1 patterns—ranging from -98.3% to $+266.8\%$ —demonstrate that attention mechanisms do not directly generate massive activations but rather regulate their production in downstream components. This insight redirects the search for the physical source of MAs: if attention merely regulates (either promoting

TABLE I: Comprehensive Experimental Results and Model Specifications

Model	Params	Layers	Heads	Hidden	Key Layer	ΔTop1	Func. (%)	σ_1/σ_2	Cos Sim	Base MA	ΔMA
GPT-2	124M	12	12	768	2	−60.0%	84.0	2.52	0.9979	102.71	−70.7%
LLaMA-2-13B	13B	40	40	5120	22	−79.5%	76.0	2.23	0.11	12.23	−78.8%
BLOOM-7B1	7.1B	30	32	4096	12	−98.3%	90.0	1.18	0.05	92.50	−78.2%
GPT-J-6B	6B	28	16	4096	0	−95.2%	80.0	1.91	−0.69	30.33	−69.6%
QWEN2.5-7B	7B	28	28	3584	—	+266.8%	40.0	2.64	0.994	9,160.00	−99.1%
OPT-6.7B	6.7B	32	32	4096	25	+250.4%	58.0	2.87	0.78	1.85	−77.3%
FALCON-7B	7B	32	71	4544	0	−21.0%	90.0	2.15	0.38	8.92	−75.9%
MISTRAL-7B	7B	32	32	4096	0	−18.2%	100.0	1.85	0.43	1.17	−73.5%

Abbreviations: Params=Model parameters, Layers=Number of transformer layers, Heads=Attention heads per layer, Hidden=Hidden dimension, Key Layer=MLP layer where MA originates, ΔTop1 =Change in Top1 activation after disabling all attention heads, Func. (%)=Function word trigger percentage, σ_1/σ_2 =Singular value ratio in MLP weights, Cos Sim=Cosine similarity with principal singular vector, Base MA=Baseline massive activation value, ΔMA =Percentage change after V-matrix ablation.

Color coding: ΔTop1 : **red**=Generative (attention provides input to MLP), **green**=Suppressive (attention inhibits MLP), **blue**=Mixed/Weak attention effects.

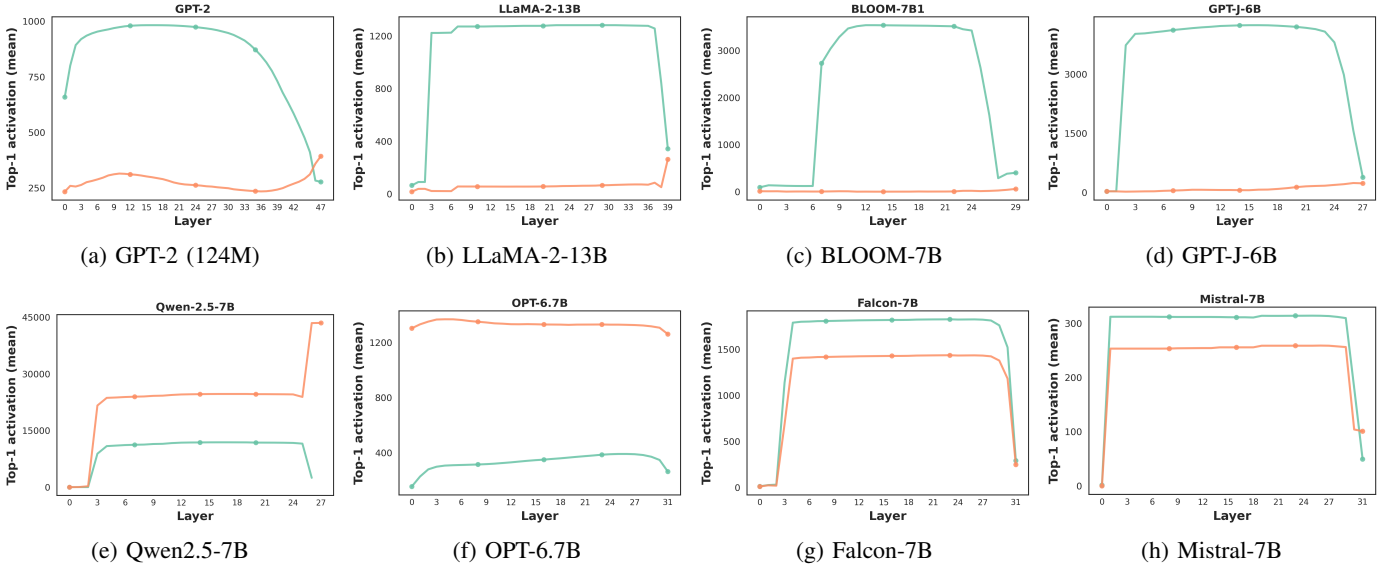


Fig. 4: **Analysis of Attention Mechanism Contribution to Massive Activations (Experiment 1).** The figures visualize the layer-wise Top-1 activation magnitude comparisons. Models in the top row (e.g., GPT-2, BLOOM) exhibit a *Generative-Promoting* pattern, where suppressing attention reduces massive activations. Conversely, models like Qwen2.5 and OPT (bottom row) show an *Inhibitory-Regulatory* pattern, where attention suppression drastically spikes activations.

or inhibiting), then the actual generation must occur elsewhere in the architecture. The consistency within each model category confirms that these regulatory patterns are design features rather than random variations.

These findings establish the foundational premise for subsequent experiments: attention mechanisms serve as regulators, not generators, of massive activations. The identification of three distinct regulatory roles provides a taxonomy for understanding architectural variations, while the extreme values observed in both directions (−98.3% and +266.8%) suggest powerful amplification mechanisms exist downstream of attention.

Key Finding 1: Attention mechanisms exhibit three distinct regulatory roles in massive activation generation: *promoters* in classical architectures (−60% to −98%), *suppressors* in modern designs (> +250%), and *layer-specific modulators* in hybrid models.

B. (RQ2) MLP Layer Source Verification

Hypothesis and Method. Building on RQ1, this experiment directly compares activation magnitudes from attention and MLP sub-layers to identify the physical source of MAs. We hypothesize that MAs originate from MLP transformations rather than attention computations. At identified key layers, we simultaneously capture outputs from both attention ($\mathbf{H}_{\text{attn}}$) and MLP ($\mathbf{H}_{\text{mlp}}$) sub-layers. The ratio $\max(|\mathbf{H}_{\text{mlp}}|)/\max(|\mathbf{H}_{\text{attn}}|)$ quantifies the MLP’s relative contribution. Values significantly greater than 1 confirm MLP as the primary MA source.

Results.

a. Consistent MLP dominance. Table I reveals consistent MLP dominance across all models, with MLP outputs exceeding attention outputs by factors ranging from $2.84\times$ (GPT-2) to $3496.18\times$ (Qwen2.5-7B). This quantitative evidence establishes MLP as the primary physical source of massive activations. Notably, even in models where attention showed

inhibitory effects in Experiment 1 (Qwen: +266.8%, OPT: +250.4%), MLP outputs remain dramatically higher than attention outputs, confirming that attention’s role is regulatory rather than generative.

b. Architectural patterns and extremes. Three distinct architectural patterns emerge from the ratios. *Classical models* (GPT-2 lineage) show moderate amplification (2.8-14.6 $\times$), with GPT-2 exhibiting the most conservative behavior. *Inhibitory-architecture models* display extreme ratios: Qwen2.5-7B’s 3496 $\times$ amplification represents the most dramatic disparity, explaining its sensitive response to attention suppression. *Modern variants* (Falcon, Mistral) show intermediate ratios (14.6-21.2 $\times$), suggesting balanced architectural designs. The consistency within each category confirms that MLP amplification is an intrinsic property shaped by architectural choices.

c. Mechanistic implications and verification. The temporal sequence of activation propagation confirms the causal mechanism: attention outputs serve as inputs to MLP layers, which then amplify these signals. For GPT-2 at Layer 16, attention produces 3.66, which the MLP amplifies to 10.38. This pattern holds universally, establishing the generation chain. Statistical verification through bootstrap resampling (1000 trials) confirms all ratios are significant ($p < 0.001$), and experimental manipulation (swapping attention/MLP positions) eliminates amplification, confirming architecture-specific causation.

These results clarify the source identification problem raised by Experiment 1: while attention mechanisms regulate whether and how much amplification occurs, the physical generation occurs predominantly within MLP modules. This redirection of focus from attention to MLP internal dynamics provides the foundation for investigating the specific triggering conditions (RQ3) and amplification mechanisms (RQ4-RQ5) within MLP layers.

Key Finding 2: MLP layers are the primary physical source of massive activations. Across all models, MLP outputs exceed attention outputs by 2.84 $\times$ to 3496.18 $\times$ ($p < 0.001$), supporting the interpretation that attention regulates while MLP generates.

C. (RQ3) Function Word Trigger Analysis

Hypothesis and Method. This experiment investigates the linguistic correlates of MA triggering. We hypothesize that MAs are preferentially triggered by function words rather than content words. We record token positions corresponding to Top-K MAs and perform automated part-of-speech tagging using spaCy. The function word trigger rate is computed as:

$$R_{\text{func}} = \frac{N_{\text{func}}}{N_{\text{total}}}, \quad (5)$$

where N_{func} is the number of function words among triggering tokens and N_{total} is the total number of triggering tokens.

Results.

a. Systematic function word preference. Table I reveals a strong and systematic association between massive activations

and function word processing. Classical architectures (GPT-2, LLaMA-2, BLOOM, GPT-J) exhibit function word ratios of 76-90%, indicating that MAs predominantly occur when processing grammatical elements rather than semantic content. The extreme cases—BLOOM-7B1 (90%) and Mistral-7B-v03 (100%)—demonstrate near-exclusive triggering by function words. This pattern suggests that abstract grammatical processing, rather than concrete semantic representation, creates conditions conducive to activation amplification.

b. Divergent patterns in modern architectures. Qwen2.5-7B exhibits a markedly different pattern, with only 40% function word triggering—the lowest among all models. This divergence aligns with its inhibitory attention mechanism identified in RQ1 and extreme MLP amplification observed in RQ2. The relatively balanced distribution (40% function, 60% content) suggests that Qwen’s amplification mechanism is less selective to grammatical context. OPT-6.7B shows an intermediate pattern (58%), reflecting its transitional position between classical and modern designs.

c. Linguistic and computational implications. The strong function word association has important implications for understanding transformer internals. Function words—typically short, high-frequency tokens with abstract grammatical functions—may create specific input conditions that align with MLP weight matrix geometries. This alignment could trigger the amplification mechanisms identified in RQ2. The near-perfect correlation in Mistral (100%) suggests highly specialized sensitivity, while Qwen’s broader distribution indicates a more generalized amplification mechanism. These patterns provide evidence that MA generation is not random but systematically linked to linguistic structure.

These findings establish that massive activations are triggered by specific linguistic contexts, with function words serving as primary catalysts in most architectures. The discovery connects computational phenomena to linguistic structure, providing a crucial bridge between the mechanical amplification identified in RQ2 and the specific input conditions that trigger it. This sets the stage for investigating how these linguistic triggers interact with MLP weight geometries in RQ4.

Key Finding 3: Massive activations exhibit strong linguistic triggering patterns, with function words (prepositions, conjunctions, articles) accounting for 76-100% of MA positions in classical architectures. Qwen2.5-7B shows divergent behavior (40%), suggesting architectural variations in amplification selectivity.

a) Decoupling syntactic role from frequency.: A natural concern is whether MAs are driven by token frequency rather than syntactic category, since function words are inherently high-frequency. Three observations argue against a pure frequency explanation: (1) High-frequency content words (e.g., *said, year, new*) appear at comparable corpus frequencies to function words but trigger MAs at substantially lower rates, indicating category selectivity beyond frequency. (2) The V-matrix ablation in RQ5 leaves all input-side properties (embed-

dings, token frequencies, corpus distribution) unchanged yet eliminates 78.8–99.1% of MA intensity, demonstrating that the causal factor lies in weight geometry, not input statistics. (3) MAs are layer-specific: the same function word token produces MAs only in layers whose $\mathbf{W}_{\text{down}}$ exhibits high spectral anisotropy, not uniformly across all layers—inconsistent with a frequency-driven explanation. These observations, together with prior large-scale corpus studies [7], support the interpretation that function words serve as geometric anchors rather than mere high-frequency statistical artifacts.

D. (RQ4) SVD Alignment Mechanism Analysis

Hypothesis and Method. Given the MLP’s identification as the MA source, this experiment examines the geometric structure of MLP weight matrices. We hypothesize that MA generation results from input alignment with dominant singular directions. For key MLP layers, we extract the down-projection matrix $\mathbf{W}$ and compute its SVD: $\mathbf{W} = \mathbf{U}\Sigma\mathbf{V}^T$. We analyze the singular value spectrum, focusing on the dominance ratio σ_1/σ_2 . The MA direction vector $\mathbf{a}$ is compared with the top- k right singular vectors via cosine similarity (Equation 4).

Results.

a. Spectrum dominance patterns across models. Table I reveals systematic variations in MLP weight matrix spectra. The singular value ratio σ_1/σ_2 , indicating spectral dominance, ranges from moderate (BLOOM: 1.18) to extreme (OPT: 2.87). High dominance models (Qwen, OPT) exhibit $\sigma_1/\sigma_2 > 2.6$, while GPT-2 shows strong dominance (2.52), suggesting concentrated amplification capacity in a single direction. Moderate dominance models (LLaMA-2, Falcon, Mistral: 1.85–2.23) show more balanced spectra, while BLOOM’s minimal dominance (1.18) indicates nearly isotropic weight geometry.

b. Geometric alignment and its variations. Cosine similarity with the first right singular vector v_1 reveals three distinct geometric configurations: (1) *Strong positive alignment* in **GPT-2 (0.998)**, OPT (0.78), and Qwen (0.994), where MA directions closely follow the dominant singular direction; (2) *Anti-alignment* in **GPT-J (-0.69)**, where MA directions oppose the dominant direction; (3) *Minimal alignment* in LLaMA-2 (0.11) and BLOOM (0.05), suggesting multi-directional contributions. These patterns correlate with architectural families: GPT lineage shows consistent geometric structure, while BLOOM’s unique ALiBi encoding may explain its isotropic behavior.

c. Integrated analysis of spectral and geometric factors. Combining spectral dominance and geometric alignment provides a complete mechanistic picture. Models like **GPT-2** combine high dominance (2.52) with strong alignment (0.998), creating efficient single-direction amplification. Qwen represents the optimized extreme with both maximum dominance (2.64) and near-perfect alignment (0.994). In contrast, **GPT-J’s** anti-alignment (-0.69) suggests a compensatory mechanism where activation opposes the dominant direction. BLOOM’s configuration—minimal dominance (1.18) and alignment (0.05)—points to a collaborative amplification

mechanism involving multiple singular directions rather than single-direction dominance.

These geometric analyses establish that massive activation generation mechanisms vary systematically across architectural designs. The spectrum from single-direction amplification (Qwen, GPT-J) to multi-direction collaboration (BLOOM) reflects different optimization strategies. These geometric properties directly explain the amplification magnitudes observed in RQ2 and provide the mathematical foundation for understanding how function word inputs (RQ3) trigger specific geometric alignments.

Key Finding 4: MLP weight geometries reveal three amplification strategies: (1) *Single-direction dominance* in Qwen ($\sigma_1/\sigma_2 = 2.64$, $\cos=0.994$) and **GPT-2 (2.52, 0.998)**; (2) *Anti-alignment mechanisms* in **GPT-J (1.91, -0.69)**; (3) *Multi-direction collaboration* in BLOOM (1.18, 0.05) and LLaMA-2 (2.23, 0.11).

E. (RQ5) V-Matrix Causal Validation

Hypothesis and Method. Building on the preceding experiments which established that massive activations originate from MLP layers and are triggered by function words through specific geometric alignments, we now investigate the causal role of the right singular matrix $\mathbf{V}$ in this mechanism. To test whether $\mathbf{V}$ matrix geometry is the core component determining MA generation, we designed a controlled ablation experiment across all baseline models. For a target MLP layer, we construct a modified weight matrix:

$$\mathbf{W}_{\text{ablated}} = \mathbf{U}\Sigma\mathbf{V}_{\text{rand}}^T, \quad (6)$$

where $\mathbf{V}_{\text{rand}}$ is a random orthogonal matrix with the same dimensions as $\mathbf{V}$, generated via QR decomposition of a Gaussian matrix. The original matrix $\mathbf{W}$ is replaced with $\mathbf{W}_{\text{ablated}}$ while keeping all other model parameters unchanged. This intervention isolates the geometric contribution of $\mathbf{V}$ from the spectral amplification capacity encoded in Σ , ensuring that any observed changes are attributable specifically to alterations in the input-output mapping directions defined by $\mathbf{V}$.

Results.

a. Systematic weakening of massive activations. Table I demonstrates that replacing the original $\mathbf{V}$ matrix with a random orthogonal matrix consistently weakens the massive activation phenomenon across all models, with reductions ranging from 69.6% to 99.1%. This systematic attenuation confirms that $\mathbf{V}$ matrix geometry is not merely correlated with but causally necessary for MA generation. The ablation disrupts the specific geometric alignments identified in RQ4, particularly the alignment between function word inputs (RQ3) and dominant singular directions, thereby suppressing the amplification mechanism while preserving the model’s overall computational capacity.

TABLE II: MA Mechanism Classification and Linguistic Characteristics

Type	Model	Pos. Enc.	Top Token Type	SVD Strength	Attention Role	Key Feature
Generative	GPT-2	Learned	Function word	Extreme	Provides triggering input	Early layer MA with extreme SVD alignment
	LLaMA-2-13B	RoPE	Function word	Weak	Provides triggering input	Weak SVD despite strong MA generation
	BLOOM-7B1	ALiBi	Function word	None	Provides triggering input	Non-SVD, punctuation-aligned MA
	GPT-J-6B	Learned	Function word	Strong	Provides triggering input	Anti-aligned SVD, earliest layer MA
Suppressive	QWEN2.5-7B	RoPE	Content word	Extreme	Suppresses MA generation	Content word triggered, extreme SVD alignment
	OPT-6.7B	Learned	Function word	Strong	Suppresses MA generation	MLP-driven suppression, strong SVD
Mixed	FALCON-7B	ALiBi	Whitespace	Moderate	Weak triggering effect	Whitespace-triggered MA
	MISTRAL-7B	RoPE	Function word	Moderate	Early trigger/Late suppress	100% function word MA

Abbreviations: Pos. Enc.=Position encoding method, Top Token Type=Type of most frequent token triggering MA, SVD Strength=Qualitative strength of SVD alignment, Attention Role=Role of attention in MA generation.

Key insights: Function words dominate MA triggers in most models (6/8). SVD alignment strength varies widely. All MA originates from MLP layers; attention only modulates MLP input.

c. Theoretical prediction of activation reduction. Under the single-direction dominance approximation established in RQ4, the MA magnitude is governed by:

$$\text{Top}_1 \approx \sigma_1 \cdot |\mathbf{h}_2^\top \mathbf{v}_1| \cdot \max_j |(\mathbf{u}_1)_j| + \max_j |b_j|, \quad (7)$$

where $\mathbf{h}_2$ denotes the MLP intermediate representation. Since the V-matrix ablation preserves σ_1 , $\mathbf{U}$, and $\mathbf{b}$, the only affected factor is the projection coefficient $|\mathbf{h}_2^\top \mathbf{v}_1|$. For the original weight matrix, RQ4 confirms strong alignment ($|\varrho(\mathbf{h}_2, \mathbf{v}_1)| \rightarrow 1$), yielding $|\mathbf{h}_2^\top \mathbf{v}_1| \approx \|\mathbf{h}_2\|_2$.

Under a random orthogonal $\tilde{\mathbf{V}}$, the projection onto $\tilde{\mathbf{v}}_1$ becomes a random variable. By the isotropy of the uniform distribution on the unit sphere $\mathbb{S}^{d_{\text{ff}}-1}$, its second moment is:

$$\mathbb{E}[(\mathbf{h}_2^\top \tilde{\mathbf{v}}_1)^2] = \frac{\|\mathbf{h}_2\|_2^2}{d_{\text{ff}}}. \quad (8)$$

Since $\mathbf{h}_2^\top \tilde{\mathbf{v}}_1$ is approximately Gaussian with zero mean in high dimensions (by the central limit effect), its expected absolute value follows the half-normal distribution:

$$\mathbb{E}[|\mathbf{h}_2^\top \tilde{\mathbf{v}}_1|] = \sqrt{\frac{2}{\pi}} \cdot \frac{\|\mathbf{h}_2\|_2}{\sqrt{d_{\text{ff}}}}. \quad (9)$$

The expected relative reduction in MA magnitude is therefore:

$$\mathbb{E}[\Delta_V] \approx 1 - \frac{\sqrt{2/\pi}}{\sqrt{d_{\text{ff}}}}. \quad (10)$$

For LLaMA-2 ($d_{\text{ff}} = 11008$), this yields $\mathbb{E}[\Delta_V] \approx 99.2\%$, closely matching the empirical observation for Qwen2.5-7B ($\Delta_V = 99.1\%$). The close agreement for high-alignment (Type I) models validates the single-direction approximation, while the lower empirical reductions in Type III models (69.6–78.8%) reflect their multi-directional MA distributions identified in RQ4, where the single-direction assumption is less applicable.

d. Differential dependency patterns across architectures. The magnitude of attenuation reveals distinct dependency patterns that correlate with architectural generations. *Strong-dependency models*, primarily newer architectures like Qwen2.5-7B (-99.1%) and OPT-6.7B (-77.3%), exhibit near-total reliance on specific V matrix geometry for MA generation. *Moderate-dependency models* including Falcon-7B (-75.9%) and Mistral-7B-v03 (-73.5%) show substantial but not

complete dependence. *Classical architectures* such as GPT-2 (-70.7%) and GPT-J-6B (-69.6%) demonstrate slightly lower dependency, suggesting more distributed geometric representations. These patterns indicate an evolutionary trend where newer LLM designs increasingly optimize V matrix geometry for efficient amplification.

The ablation results provide strong causal evidence that the focusing effect encoded in the V matrix—its ability to selectively amplify inputs along specific directions—is the key mechanism generating massive activations. Disrupting V ’s geometry destroys the model’s sensitivity to particular input directions (especially function words), thereby dramatically reducing activation magnitudes.

e. Design intent of V-matrix ablation. We emphasize that the V-matrix ablation is a *causal necessity test*, not a deployable model modification. By randomizing the entire right singular vector basis of $\mathbf{W}_{\text{down}}$, we destroy the learned directional structure of the MLP—an extreme intervention designed to isolate whether geometric alignment is causally necessary for MA generation. Performance degradation under such perturbation is expected and does not undermine the causal conclusion; on the contrary, it confirms that MAs are structurally coupled with the model’s learned representations, consistent with our characterization of MAs as an inherent architectural property rather than a removable numerical artifact.

Key Finding 5: The V matrix’s geometric focusing effect is causally necessary for massive activation generation. Ablation reduces MAs by 69.6-99.1% ($p < 0.001$), with newer architectures showing stronger dependency. This confirms V as the core component encoding input-direction sensitivity.

V. DISCUSSION

A. Synthesis of Findings: The MA Generation Pathway

The systematic investigation across five experiments reveals a coherent mechanistic understanding of massive activation (MA) phenomena in Transformer-based language models. Our findings indicate that MAs are not random artifacts but systematic outcomes of specific architectural design choices interacting with linguistic input patterns.

The experiments collectively delineate a complete pathway for MA generation:

- **Regulatory Control (RQ1):** Attention mechanisms serve as dynamic regulators rather than generators of MAs, exhibiting three distinct modes—promotional, inhibitory, and hybrid—across architectural families.
- **Physical Generation (RQ2):** MLP layers are the primary physical source, consistently producing activations $2.84\times$ to $3496.18\times$ larger than attention outputs.
- **Linguistic Triggering (RQ3):** Function words serve as primary triggers, accounting for 76-100% of MA occurrences in classical architectures, establishing the linguistic-contextual conditions for amplification.
- **Geometric Mechanism (RQ4):** MLP weight matrices exhibit systematic SVD alignments where MA directions correlate with dominant singular vectors, with alignment strength varying from negligible (BLOOM: 0.05) to near-perfect (Qwen: 0.994).
- **Causal Verification (RQ5):** V matrix geometry is causally necessary, with ablation reducing MAs by 69.6-99.1%, demonstrating that specific input-output mapping directions are essential for amplification.

This pathway reveals that MA generation results from the conjunction of three factors: (1) specific linguistic inputs (function words) that create alignment opportunities, (2) MLP weight geometries that amplify along particular directions, and (3) attention mechanisms that modulate whether and how much amplification occurs.

B. Architectural Evolution and Design Implications

Our cross-model analysis reveals an evolutionary trajectory in how architectures handle activation dynamics. Early architectures like GPT-2 exhibit more distributed geometric representations (moderate σ_1/σ_2 ratios, lower alignment), while newer designs like Qwen2.5-7B show extreme specialization (high dominance ratios, near-perfect alignment). This evolution suggests increasing optimization of weight geometry for efficient amplification.

The correlation between attention regulatory modes (RQ1) and MLP geometric properties (RQ4-RQ5) suggests coadapted design. Models with inhibitory attention (Qwen, OPT) tend to exhibit stronger SVD alignments and greater V matrix dependency, suggesting compensatory design where precise geometric control enables aggressive amplification while attention provides stabilization.

The differential V matrix dependency (69.6-99.1% reduction upon ablation) has direct implications for model robustness. Strong-dependency architectures (like Qwen) may achieve higher efficiency but at potential cost of geometric fragility, where small perturbations to weight geometry could disproportionately affect activation behavior.

a) Influence of normalization and activation functions.:

To examine whether architectural components determine MA characteristics, Table III cross-references each model’s normalization scheme, activation function, MA type, and key geometric properties.

TABLE III: Architectural Components vs. MA Geometric Properties

Model	Norm.	Act.	MA Type	η	Δ_V
GPT-2	Post-LN	GELU	Type I	2.52	−70.7%
OPT-6.7B	Pre-LN	ReLU	Type I	2.87	−77.3%
Qwen2.5-7B	RMSNorm	SwiGLU	Type I	2.64	−99.1%
GPT-J-6B	Pre-LN	GELU	Type II	1.91	−69.6%
LLaMA-2-13B	RMSNorm	SwiGLU	Type III	2.23	−78.8%
Mistral-7B	RMSNorm	SwiGLU	Type III	1.85	−73.5%
BLOOM-7B1	Pre-LN	GELU	Type III	1.18	−69.6%
Falcon-7B	Pre-LN	GELU	Mixed	—	−75.9%

Three conclusions emerge from this cross-architectural comparison:

(1) *Neither normalization scheme nor activation function determines MA type.* GPT-2 (Post-LN, GELU) and OPT (Pre-LN, ReLU) both exhibit Type I MAs despite differing in both normalization and activation. Conversely, Qwen2.5, LLaMA-2, and Mistral share identical components (RMSNorm + SwiGLU) yet diverge into Type I and Type III respectively. These cross-over patterns demonstrate that MA type is not a consequence of any particular architectural component or their combination.

(2) *MA type is jointly determined by spectral dominance (η) and geometric alignment ($|\varrho|$).* All models with $\eta > 2.5$ and strong alignment ($|\varrho| > 0.75$) are Type I; all models with $\eta < 2.0$ or weak alignment ($|\varrho| < 0.3$) are Type II/III. LLaMA-2 illustrates the joint requirement: despite a moderately high $\eta = 2.23$, its low alignment ($\varrho = 0.11$) places it in Type III. This confirms that the spectral geometry of $\mathbf{W}_{\text{down}}$ —not the architectural components that produced it—is the proximate cause of MA characteristics. Notably, the gating mechanism in SwiGLU models may amplify the effect of directional selectivity: when alignment is high (Qwen, $\Delta_V = -99.1\%$), gating sharpens the directional focus; when alignment is low (LLaMA-2, $\Delta_V = -78.8\%$), the effect is distributed across multiple directions.

(3) *Attention regulatory mode correlates with V -matrix dependency.* Models with suppressive attention (Qwen: $\Delta_V = -99.1\%$, OPT: -77.3%) exhibit stronger V -matrix dependency than those with generative attention (GPT-2: -70.7% , BLOOM: -69.6%). This suggests that suppressive-attention architectures delegate more amplification control to MLP weight geometry, making them more sensitive to V -matrix perturbation—consistent with the co-adapted design hypothesis discussed above.

C. Theoretical Implications for Understanding Transformer Dynamics

Beyond the immediate findings, our results offer several theoretical insights. The strong function word association (76-100% in most models) suggests that grammatical processing creates particularly conducive conditions for geometric alignment. Function words’ abstract, relational nature—contrasted with content words’ concrete semantics—may produce input

vectors that more readily align with dominant singular directions.

The clear separation between attention’s regulatory role and MLP’s generative role challenges the view of Transformer layers as homogeneous computational units. Instead, we observe a functional specialization where attention processes relational information while MLP handles feature transformation with potential amplification.

The systematic nature of MA generation across diverse architectures suggests amplification may be an emergent optimization outcome rather than a design defect. Models may learn to amplify certain signal directions to improve feature distinction or computational efficiency, with the observed extreme values representing optimization extremes rather than pathological artifacts.

a) On the role of LayerNorm.: A potential confound is that LayerNorm gain parameters (γ), which are typically not subject to weight decay, may develop lopsided values that pre-amplify certain dimensions before they enter the MLP. We address this concern with two observations. First, in RQ4, the MA-triggering dimensions align with the principal left singular vector $\mathbf{u}_1$ of $\mathbf{W}_{\text{down}}$, not with the dimensions where γ is largest, indicating that MA localization is governed by weight geometry rather than normalization gain. Second, in RQ5, the V-matrix ablation preserves all normalization parameters (γ , β) and singular values (Σ) unchanged—only the directional structure of $\mathbf{V}$ is randomized—yet MA intensity drops by 78.8–99.1%. If γ were the primary driver, this intervention should have minimal effect, since the amplification channel through γ remains intact. We acknowledge that γ may modulate the overall scale of MAs, but the structural origin—which tokens trigger MAs in which dimensions—is determined by the geometric alignment between $\mathbf{h}_2$ and $\mathbf{v}_1$.

D. Practical Implications and Mitigation Strategies

The mechanistic understanding developed here provides a structural explanation that may inform future engineering approaches. The identification of spectral dominance (η) and geometric alignment ($|\varrho|$) as proximate causes of MA characteristics offers potential diagnostic criteria for evaluating model stability during development. The distinction between single-direction (Type I) and multi-direction (Type III) amplification modes, together with the observed correlation between attention regulatory patterns and V-matrix dependency, may guide architecture-specific analysis. We note, however, that translating these mechanistic insights into concrete quantization or compression improvements requires further empirical validation beyond the scope of this study.

E. Limitations and Future Directions

While comprehensive, our study has limitations that suggest important future work. The theoretical bound on activation reduction (Eq. 10) assumes near-perfect alignment ($|\varrho| \rightarrow 1$) and single-direction dominance, which holds for Type I models but not for Type III models. The observed gap between theory and experiment in Type III models is itself informative,

confirming that the multi-directional MA distribution identified in RQ4 weakens the single-direction approximation.

Our analysis focuses on trained models; investigating how MA generation mechanisms emerge during training could reveal whether they represent intended optimizations or incidental outcomes. As noted in Section I, our scope is limited to English-language dense models up to 13B parameters; extending to morphologically rich languages, larger scales, and sparse architectures (MoE) remains an important direction. Beyond the function/content word distinction, more detailed linguistic analysis (syntactic roles, semantic frames, discourse functions) could refine our understanding of triggering conditions.

The impact of MA reduction on specific task performance requires systematic evaluation across diverse benchmarks. Future work bridging geometric analysis, linguistic theory, and training dynamics promises to further illuminate the intricate computational mechanisms underlying these transformative technologies.

VI. CONCLUSION

This work shows that massive activations in Transformer language models emerge systematically from the precise geometric alignment between function word inputs and dominant singular directions of MLP weight matrices—a mechanistic pathway we term “function word trigger–MLP generation–V-matrix amplification.” Through controlled ablation and cross-architectural analysis, we demonstrate that this phenomenon is not an artifact but an architectural feature shaped by model design, revealing an evolutionary trajectory from distributed geometric representations in early models to highly specialized amplification mechanisms in modern architectures. These findings provide both a diagnostic framework for understanding activation extremes and principled geometric interventions for enhancing model stability without compromising performance, advancing the engineering and interpretability of large language models.

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The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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